

Insider Threat

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Business brief

You have been tasked with reducing alert fatigue for security analysts at a large tech company. Their insider threat stack is mature, with multiple models, IDS-based solutions, and threat hunting methods integrated into providing graph-based events. However, work is still needed to triage, present and explain the events that the analyst sees. Treat each subgraph as the atomic element that needs to be scored.

Data

You have one file to work with:

- security_data.json

Your JSON contains three arrays:

- events: Security alerts with timestamps, severity scores, and descriptions
- entities: Users, hosts, files, processes, domains, databases, network connections
- relationships: Links between events and entities (event_id -> entity_id)

You will need to construct the graphs yourself in order to properly process this data.

Project objectives

- Develop a method to score and rank each graph event in the data.
- Map these events to the MITRE ATT&CK framework and use that framework to ensure that temporal logic is employed in the ranking of the event graphs.
- Determine and implement a threshold that determines which events should be presented to the analyst.

- Deliver the event rankings with any auxiliary information you find helpful on a dashboard for the analyst to consume.
- Provide a method for the analyst to tag events appropriately, in a standardized fashion. These labels can be used later to retrain the model.

Evaluation

Please prepare a presentation of your findings. Ensure that you present:

1. The business needs
2. Your methodology
3. The results including any KPIs that reflect the objective of this project.

In addition return both the code and any output files for evaluation. Include instructions for installation and operation. Ensure that your code is ready to receive new input files for stress testing.

Submission

Presentation, code and output files must be submitted by email midday, Monday June 14th. Keep in mind the file size constraints of email, so be ready to upload to google drive.